Radon Monitoring

User manual

Local monitoring for GQ RadonScan devices

Version	4.9.0
Date	26 July 2026
Home Assistant	Local app / Ingress
Data	Local SQLite + optional external upload

Contents

Important: this app is a monitoring and documentation tool, not an official measurement system.

1. Purpose, measurement principle and limitations

Radon Monitoring reads completed hourly records from compatible GQ RadonScan devices using read-only SPIR commands. It stores raw counts (CPH), the conversion factor used for each record and the calculated activity concentration locally in SQLite.

The app supports continuous observation, statistical analysis and traceable documentation. It does not replace traceable calibration, a recognised regulatory measurement or professional medical, building or legal advice.

A high individual hourly value is not automatically an exceedance of an annual reference value. The interface and reports distinguish the current value, period statistics, counting uncertainty, calibration information and professional interpretation.

2. Installation, upgrade and first start

Connect the RadonScan by USB, make MQTT available to Home Assistant and start the app. Leave serial_port empty or set to auto for discovery. For a stable fixed assignment, prefer a path under /dev/serial/by-id/.

Create a Home Assistant backup before upgrades and destructive actions. After upgrading, the sidebar or Help view must show version 4.9.0. If an old Ingress view remains open, close the panel and reopen it.

Only completed hours are imported. A newly connected device can therefore remain without a current value until a complete record is available.

3. Overview

Overview shows the latest completed hourly value, data age, raw value, measurement site, stored hours, device connection and MQTT state. It also shows 24-hour, 7-day and 30-day summaries.

A period result is displayed only after the required duration and minimum coverage have been reached. Otherwise the card explains why the value is not yet meaningful.

Refresh loads the compact system state first. History, catalogue and analysis follow separately so that slower requests do not hide newly imported USB data.

4. Analysis and statistics

Basic analysis includes mean, median, minimum, maximum, quantiles, coverage, quality class, threshold time and robust dispersion measures. Advanced analysis adds effective sample size, moving-block-bootstrap confidence intervals, Mann-Kendall diagnostics, Sen slope, autocorrelation, distribution diagnostics, rolling median, interquartile band and continuous threshold events.

Missing measurements are not imputed. Unusual observations are not deleted automatically. Quality flags identify undocumented factors, non-primary sources, implausible values and duplicate timestamps.

Change-point analysis is exploratory. For very long series, only this diagnostic is bounded to an evenly distributed, time-ordered sample. Descriptive statistics, threshold calculations and data checksums continue to use every selected record.

Statistical significance proves neither causality nor practical importance. Events and interventions still require professional interpretation.

5. Expert view, factor and calibration

Expert view shows firmware, serial port, decoder, SPIR checksums, database state, raw counts, runtime diagnostics and factor history.

The currently configured factor is displayed separately from the factor stored with the selected historical measurement. Historical results therefore remain reproducible after later configuration changes.

Calibration records can include date, laboratory, certificate reference, factor, uncertainty, next due date and notes. A calibration entry does not silently alter existing measurements.

6. Sites, sessions and events

Measurement sites, sessions and events are local metadata. They document building, room, placement, height, ventilation, interventions, device changes and outages without changing original measurements.

For scientific comparisons, record the campaign question, start, end and relevant environmental conditions.

7. GQ Radiation World Map

Upload is optional and disabled by default. Account ID and device ID are stored in app options and shown only in masked form in the interface.

A persistent queue prevents duplicates, retries temporary failures with increasing delay and can discard measurements older than a configured limit. Manual upload is a protected POST action.

The public position is managed in the GQ account. Radon Monitoring does not transmit its own GPS coordinates.

8. Scientific PDF reports

Compact and detailed reports contain period, site, device, factor, coverage, statistics, time series, method, quality information, software version and SHA-256 checksums.

For multi-year series, only the time-series chart is reduced to a bounded peak-preserving point set. Statistics, threshold events and the data checksum continue to use all selected measurements.

Reports are documentation aids, not official certificates.

9. Data management and security

The complete Data management area can be hidden from navigation and blocked at the API using data_management_enabled.

Complete local reset automatically creates a safety backup, clears app data transactionally, verifies SQLite integrity and reports operation ID, deleted records, removed files, sizes and duration. Immediate re-import of the deleted device history is suppressed.

Home Assistant Recorder purge submits recognised RadonScan entities and stable name patterns to recorder.purge_entities. An optional long-lived administrator token may be required. It is never displayed, excluded from diagnostics and centrally redacted from error messages.

Verification subsequently checks recent Home Assistant history. It does not prove immediate physical compaction of every Recorder backend and does not automatically cover separately retained long-term statistics.

10. Operation, mobile layout and accessibility

On small screens the sidebar opens as a menu. Tables and the weekly heatmap remain horizontally scrollable inside their own cards; the rest of the page must not create horizontal page overflow.

The interface supports keyboard operation, visible focus, a skip link, Escape to close the menu, reduced motion and text scaling to 200 percent.

Version 4.9.0 was checked in Chromium at 320, 390, 768 and 1440 pixels, in all eight languages and with deliberately long labels.

11. Troubleshooting

No device: check USB mapping, permissions, configured port and competing processes.

No new values: only completed hours are imported. Press Refresh and inspect the latest scan time in Expert view.

Old interface: restart the app, close the Ingress panel and reopen it.

Recorder purge fails: check administrator token, Recorder integration and Home Assistant administrator rights.

Verification reports remaining data: wait and verify again; Recorder maintenance can be asynchronous. Treat long-term statistics separately.

Restore rejected: only readable SQLite backups with a compatible schema are accepted. The active database remains unchanged after validation failure.

12. Privacy and responsible use

Measurements and metadata remain local by default. Only enabled external functions transmit data. Before World Map publication, confirm that location and publication settings meet your privacy requirements.

Protect backups and reports because they may contain device identifiers, sites, periods and building information.

13. New in version 4.9.0

Modular backend and frontend data management; central token redaction; Recorder purge verification; safe browser storage; corrected World Map POST route; complete delivery of split JavaScript assets; USB reconnect tests; fast multi-year reports; root test configuration; native option translations and expanded browser end-to-end tests.

Configuration overview

Option	Purpose
serial_port	Automatic discovery or stable USB path
factor_bq_m3_per_cph	Currently configured conversion factor
minimum_data_coverage_percent	Minimum coverage for period results and quality
analysis_timezone	Time zone for daily and weekly profiles
data_management_enabled	Enable Data management navigation and API
homeassistant_access_token	Optional administrator token for Recorder actions
gmcmap_enabled	Enable GQ Radiation World Map
gmcmap_auto_upload	Enable the automatic upload queue

Further technical details are available in DOCS.md and the built-in Help view.